FILE PERMISSIONS IN LINUX

3.LINK COUNT

Link count is the total number of names referring to the same file data.

Link counts are:

1. File----------> (1)
2. Directory----------> (2)
3. Parent directory----> (3)

Note:

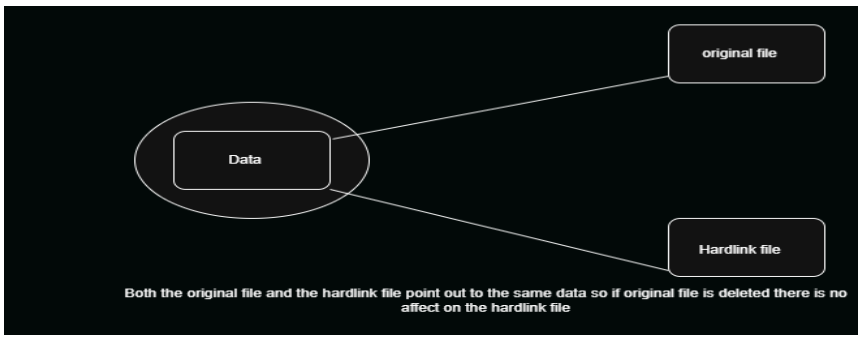
- Directory link count is not always 2 or 3
- It increases when subdirectories are created

HARDLINK: - (ln)

1.These links have the actual file content (original file content) but it points out to the data which is stored it on the disk
2.link count increases by 1
3.inode number is same (ls -i)
4.cannnot create hardlink of directory
5.if u write anything on the original file it will automatically replicate to the hardlink file
6.its file type is normal file.

example:

ln file1 file2 ----------> i have created the file2 as a hardlink file for file1

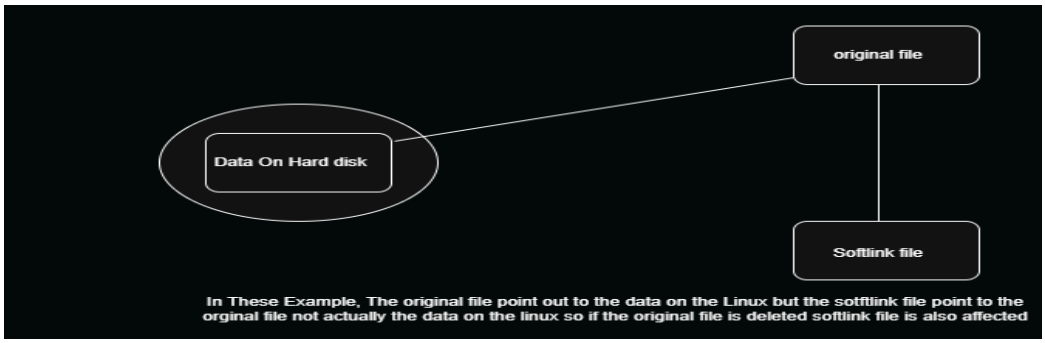


SOFTLINK: (ln -s)

1. Used to create a shortcut file (pointer) to another file
2.link count does not increase
3.inode number is different (ls -i)
4.can create softlink of file as well as directory

5.if u delete the original file, softlink file colour changes from blue to red
6.its file type is link file.

example: -
ln -s file1 file2 ----------> i have created the file2 as a shortcut file for file1



INODE VALUE (number): -

1.inode simply means a index node a uniq number assigned to every file in Linux
2. It stores information (metadata) about the file, not the file name.
3**.example: -** ls -i filename

INODE contains:

- File size
- File permissions
- Owner (user ID)
- Group ID
- File type
- Timestamps (created, modified, accessed)
- Link count

Example: - Adhar number of a person, name change but the number remain same

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4.OWNERSHIP

ownership means basically who owns the file and the group of that file

1.To change owner: -
chown Linux file

2.To change the group: -
chgrp Linux file

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3.To change both owner and group: -

chown linux:linux file

Example: -

1.touch new

ls -l

-rw-r--r-- 1 root root 0 Apr 11 16:53 file

2.chown mayur:mayur file

ls -l

-rw-r--r-- 1 mayur mayur 0 Apr 11 16:53 file
